OR-AS Operations Research Applications and Solutions	Case Name: Apartment Building Finishing Works (4)	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	William & Jef		User defined distributions
Date	2016	Project Control	Automatic tracking
File Name	C2016-34 Apartment Building Finishing Works (4)		Tracking based on user input

1. Project description

Project authenticity

This project covers the finishing works for the +2 and +3 levels of an apartment building (Block B), planned from March 21 to July 13, 2016, with a baseline duration of 83 working days and a total budget of €362,476. Key activities include plastering, flooring, interior carpentry, painting, furniture installation, and final cleaning. Three tracking reports show the project was completed later than planned (92 actual vs. 83 baseline days) and over budget (€383,871 actual vs. €362,476 baseline).

2. Project properties

2.1. Baseline Schedule

General	
# Activities	23
Planned Duration (PD)	83
Budget At Completion (BAC)	362,476.00€
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	40%
Activity Distribution (AD)	43%
Length of Arcs (LA)	0%
Topological Float (TF)	26%

* standard eight-hour working days=

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0.00	0.00	N/A
CRI-rho	100.00	0.00	N/A
CRI-tau	100.00	0.00	N/A

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	-	-	-
CRI-rho	-	-	-
CRI-tau	-	-	-

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	31	26	0.27
SI	50	25	-0.12
SSI	13	16	1.44
CRI-r	16	15	1.28
CRI-rho	16	16	1.27
CRI-tau	10	10	1.37

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	25.24	-15.78
PV - SPI	23.32	3.09
PV - SCI	23.32	3.09
ED - 1	29.27	-14.71
ED - SPI	23.32	3.09
ED - SCI	23.32	3.09
ES - 1	23.38	-23.22
ES - SPI(t)	18.48	0.77
ES - SCI(t)	18.48	0.77

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.00	0.00
CPI	0.00	0.00
SPI	12.88	12.88
SPI(t)	15.36	14.52
SCI	12.88	12.88
SCI(t)	15.36	14.52
0.8 CPI + 0.2 SPI	4.27	4.27
0.8 CPI + 0.2 SPI(t)	5.14	5.04

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 3 tracking periods. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-18654.67	-44878.00	-7	0.91	0.81	0.88	1
std dev	2701.38	52010.84	1.73	0.03	0.18	0.04	0
final	-21395.00	0.00	-72.00	0.94	1.00	0.90	1

2.3.3.2. Time forecasting

PD	83	Real Duration	92	Late	10.84%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	129.56	17.13	40.82	-40.82
PV - SPI	146.58	34.09	59.33	-59.33
PV - SCI	160.87	39.40	74.86	-74.86
ED - 1	132.25	9.58	43.75	-43.75
ED - SPI	150.92	28.28	64.04	-64.04
ED - SCI	158.19	32.51	71.95	-71.95
ES - 1	124.04	2.89	34.83	-34.83
ES - SPI(t)	130.90	5.34	42.29	-42.29
ES - SCI(t)	136.57	10.60	48.45	-48.45

2.3.3.3. Cost forecasting

BAC	362,476.00€	Real Cost	383, 871. 00 €	Over budget	5.90%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	381130.67	2701.38	0.71	0.71
CPI	396654.58	11365.77	3.33	-3.33
SPI	435173.94	51865.19	13.36	-13.36
SPI(t)	403112.65	19374.75	5.01	-5.01
SCI	456626.20	67849.57	18.95	-18.95
SCI(t)	421149.77	35697.11	9.71	-9.71
0.8 CPI + 0.2 SPI	402728.32	16383.74	4.91	-4.91
0.8 CPI + 0.2 SPI(t)	397887.41	12765.31	3.65	-3.65